# Department of Information Technology

**FACULTY OF ENGINEERING AND TECHNOLOGY**
**UNIVERSITY OF SINDH, JAMSHORO**

## Electronic Devices and Circuits (GNAS-317)

**LAB # 01**

Name: _____ Roll No: _____

Score _____Signature of the Lab Tutor: _____ Date: _____

| **Lab Performance Indicator** | Selection of equipment (0 to 5) | Utilization of lab tools and calibration (0 to 5) | Experimental Data collection (0 to 5) | Experimental data analysis (0 to 5) | Lab safety precautions (0 to 5) | Team work during lab task (0 to 5) | Technical cleanliness and organization (0 to 5) | Score=2/35*obtained marks |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | |

# To understand the basic concepts of measurements and lab safety to avoid electric shock

## PERFORMANCE OBJECTIVE:

Upon successful completion of this experiment, the student will be able to:

- Understand the basic concepts of measurements, basic terminology.
- Understand the hazards of the electric shock on human body and rules for safe practice to avoid electric shock.
- Show how much amount of current is changed within human body with the variation of body resistance.

### EQUIPEMENT:

- DC ammeter (range 0-50µA)
- Multimeter with resistance scale
- 9-12 V battery
- AC Supply

### PRECAUTIONS:

- Take care of the equipment. They can be damaged by misuse.
- Before switch on your circuit please get it checked by your teacher.
- Don’t perform the experiment of current flowing within human body at high voltage.

**Theoretical Background:**

**MEASUREMENTS:**

Measurement is perhaps one of the most fundamental concepts in science. Without the ability to measure, it would be difficult for scientists to conduct experiments or form theories. Not only is measurement important in science and the chemical industry, it is also essential in farming, engineering, construction, manufacturing, commerce, and numerous other occupations and activities.

The word "measurement" comes from the Greek word "metron," which means "limited proportion." Measurement is a technique in which properties of an object are determined by comparing them to a standard.
Measurements require tools and provide scientists with a quantity. A quantity describes how much of something there is or how many there are. A good example of measurement is using a ruler to find the length of an object. The object is whatever you are measuring, the property you are trying to determine is the object's length, and the standard you are comparing the object's length to is the ruler.

In general, scientists use a system of measurement still commonly referred to as the "metric system." The metric system was developed in France in the 1790s and was the first standardized system of measurement. Before that time, people used a variety of measurement systems.
In 1960, the metric system was revised, simplified, and renamed the Système International d'Unites (International System of Units) or SI system (meters, kilograms, etc.). This system is the standard form of measurement in almost every country around the world, except for the United States, which uses the U.S. customary units system (inches, quarts, etc.). The SI system is, however, the standard system used by scientists worldwide, including those in the United States.

There are several properties of matter that scientists need to measure, but the most common properties are length and mass. Length is a measure of how long an object is, and mass is a measure of how much matter is in an object. Mass and length are classified as base units, meaning that they are independent of all other units. In the SI system, each unit of measure has a base unit.
The seven base units of the SI system are listed in the table below:

| Measure | Base Unit |
|---|---|
| Length | Meter (m) |
| Mass | Kilogram (kg) |
| Time | Second (s) |
| Temperature | Kelvin (K) |
| Amount of substance | Mole (mol) |

| | |
|---|---|
| Electric current | Ampere (A) |
| Luminous intensity | Candela (cd) |

**Errors:**

Errors are a measure of the lack of certainty in a value.

**Example:** The width of a piece of A4 paper is 210.0 ± 0.5 mm. I measured it with a ruler1 divided in units of 1 mm and, taking care with measurements, I estimate that I can determine lengths to about half a division, including the alignments at both ends. Here the error reflects the limited resolution of the measuring device.

**Example:** An electronic balance is used to measure the weight of drops falling from an outlet. The balance measures accurately to 0.1 mg, but different drops have weights varying by much more than this. Most of the drops weigh between 132 and 139 mg. In this case we could write that the mass of a drop is (136 ± 4) mg. Here the error reflects the variation in the population or fluctuation in the value being measured.

**Types of Errors**

**Systematic and random errors:**

A systematic error is one that is reproduced on every simple repeat of the measurement. The error may be due to a calibration error, a zero error, a technique error due to the experimenter, or due to some other cause. A random error changes on every repeat of the measurement. Random errors are due to some fluctuation or instability in the observed phenomenon, the apparatus, the measuring instrument or the experimenter.

**Independent and dependent errors:**

The diameter of a solid spherical object is 18.0 ± 0.2 mm. The volume, calculated from the usual formula, is 3.1 ± 0.1 cm3 (check this, including the error). These errors are dependent: each depends on the other. If I overestimate the diameter, I shall calculate a large value of the volume. If I measured a small volume, I would calculate a small diameter. Any measurements made with the same piece of equipment are dependent.
Suppose I measure the mass and find 13.0 ± 0.1 g. This is an independent error, because it comes from a different measurement, made with a different piece of equipment. There is a subtle point to make here: if the error is largely due to resolution error in the measurement technique, the variables mass measurement and diameter measurement will be uncorrelated: a plot of mass vs diameter will have no overall trend. If, on the other hand, the errors are due to population variation, then we expect them to be correlated: larger spheres will probably be more massive and a plot will have positive slope and thus positive correlation. Finally, if I found the mass by measuring the diameter, calculating the volume and multiplying by a value for the density, then the mass and size have inter-dependent errors.

**Absolute error:**
The error expressed in the same dimensions as the value. e.g. 43 ± 5 cm.

**Percentage error:**

The error expressed as a fraction of the value. The fraction is usually presented as a percentage. E.g. 43 cm ± 12%

**Formula:**

$$\text{Percentage error} = \left(\frac{\text{experimental value} - \text{theoretical value}}{\text{theoretical value}}\right) * 100$$

**Accuracy:**

A measure of how close the observed value is to the true value. A numerical value of accuracy is given by:

$$Accuracy = 1 - \left(\frac{Obseved\ value - true\ value}{true\ value}\right) * 100$$

**Precision:**

A measure of the detail of the value. This is often taken as the number of meaningful significant figures in the value.

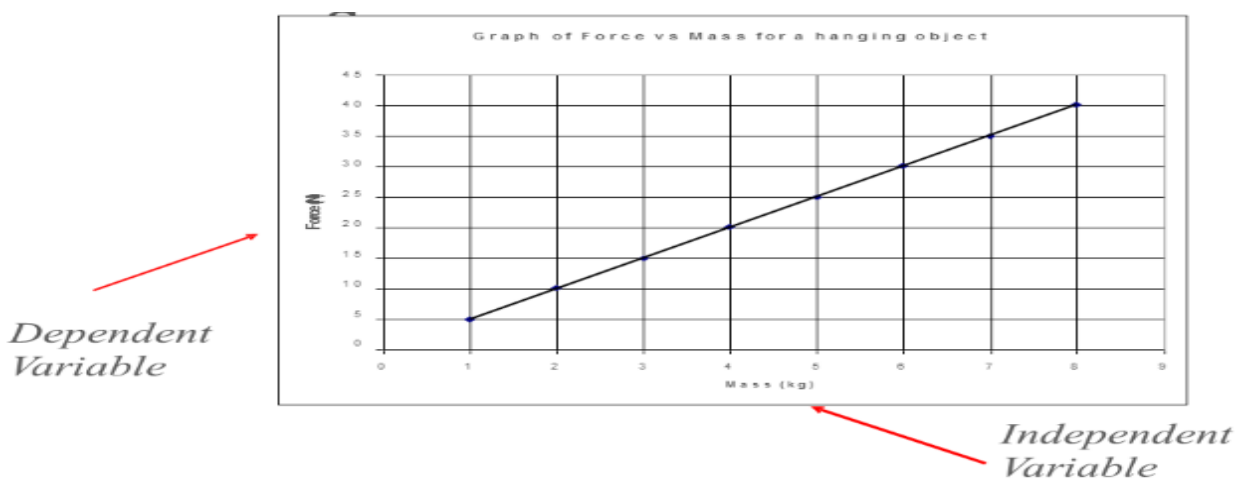
**Graphs:**

In many subject areas you may need to refer to numbers, statistics and other data during the course of your studies. This is likely to be data collected by other people which you will use to support your written work, but it may be data that you have collected yourself as part of your studies. Data is generally presented in the form of tables, charts and graphs, which makes it easier for readers to understand. However, it is often necessary to reproduce and refer to this type of information in words, as part of a report or written assignment. If you include a graph, chart or table in your writing, you must explain very clearly what the data in it means, and why it is relevant to your report or assignment.

**Linear Graphs:**

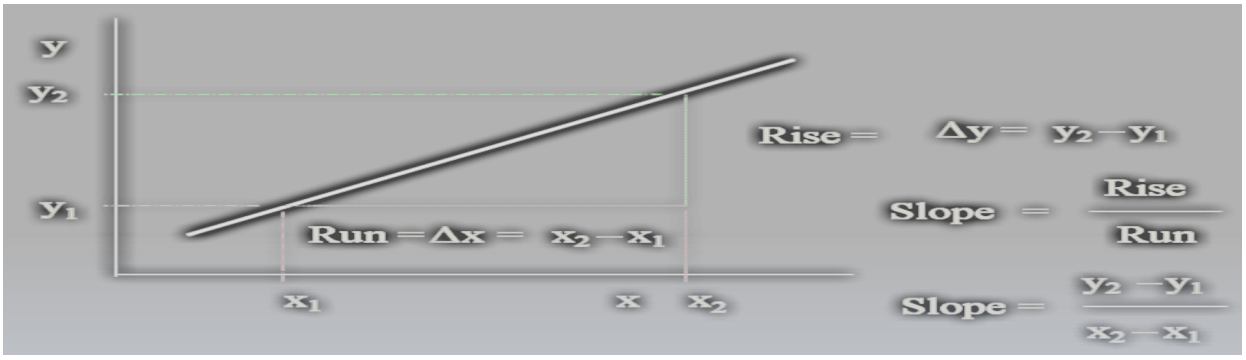
Graphing data shows if a relationship exists between two quantities also called variables.
If two variables show a linear relationship they are directly proportional to each other.

Examine the following graph:




**Linear Graphs – Slope of a Line:**

The slope of a line is a ratio between the change in the y-value and the change in the x- value.
This ratio tells whether the two quantities are related mathematically.
Calculating the slope of a line is easy.




**Linear Graphs – Equation of a Line:**
Once you know the slope then the equation of a line is very easily determined.
Slope Intercept form for any line:

$$y = mx + b$$

slope

y-intercept
(the value of y when x =0)

In Physics we don't use "x" & "y". We could use F and m, or d and t, or F and x etc.

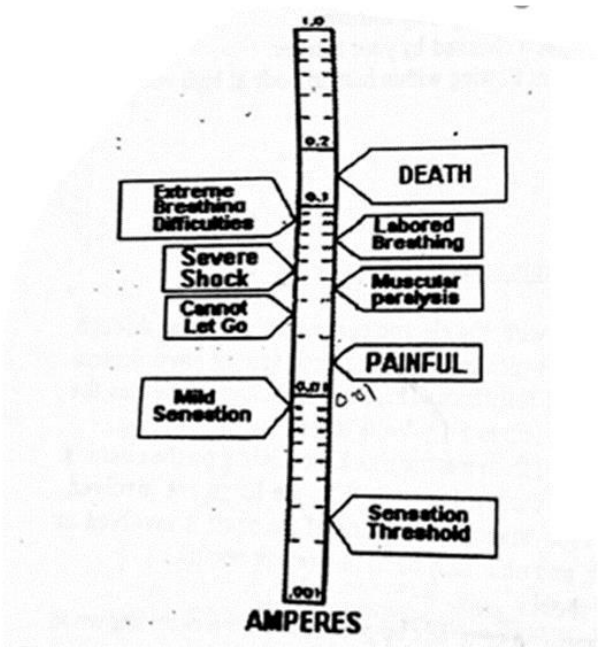
## LAB SAFETY

To understand the hazards of electric shock on human body and know the rules for safe practice to avoid an electric shock.

**Electric Shock**

**Demonstration of current flowing within human body**

When a person comes in contact with a live wire, the electric current starts flowing through the body to earth. Although, it is current which causes a shock and various physiological effects, but current depends on voltage and resistance. Therefore, as voltage increases the current also increases. In most cases, voltage up to 50 volts is safe.
The human physiological conditions can result from electric shock, depending on the circuit's route through the body and the amount of current. If the lungs are involved, voluntary respiration can halt, and asphyxiation takes place. If the heart is involved in the circuit, irregular heartbeats can occur, and ultimately heart failure can result. In extremely fatal cases, electric shock may cause death which is called electrocution.
The physiological effects of electric currents can generally be predicted by the chart shown in figure 1.1. Notice that it is the current that does the damage. Currents above 100mA or only 1/10 of an ampere are fatal. Currents below 100 mA can be serious and painful. Skin resistance may vary between 250 ohms for wet skin to 50,000 ohms for dry skin.




**Safety Rules**

The apparent reasons for accidents are:

- Carelessness
- Fatigue
- Mental pressure
- Faulty or Improper tools
- Incorrect procedure

**Rules for safe practice and to avoid electric shock:**

**1.** Be sure of the conditions of the equipment and the dangers present before working on the piece of the equipment. Many technicians are killed by supposedly "dead" circuits.

**2. Never rely on safety devices.**

Such as fuses, relays, and interlock systems to protect you. They may not be working and may fail to protect when most needed.

**3. Never remove the GROUND wire of the three-wire input plug.**

This eliminates the grounding feature of the equipment making it potential shock hazard.

**4. Do not work on cluttered bench.**

A disorganized mess of connecting leads, components and tools only leads to careless thinking, short circuits, shocks, and accidents. Developed habits of systemized procedure of work.

**5. Do not work on wet floors.**

Your contact resistance to ground is substantially reduced. Work on rubber mate or an insulated floor.

**6. Do not work alone.**

It is better to someone around to shut off the power to give artificial respiration and to call a doctor.

**7. Work with one hand behind you and in your pocket.**
A current between two hands crosses your heart and can be more fatal than a current from hand to foot. Wise technicians always work with one hand.

**8. Never talk to anyone while working.**
Do not divert your mind by talking to anyone. Also do not talk to anyone, if he/she is working on dangerous equipment.

**9. Always move slowly when working around electrical circuits.**
Violent and rapid moments lead to accidental shocks and short circuits.

**10.** If you have to modify the circuit connections, first switch off the power, take out power plug/wires and only then modify it.

## Electrical Hazards

Even the most experienced research professional can overlook basic safety principles when working with electricity – so it is vital that protective measures are employed throughout the laboratory. Particularly hazardous are electrical units which are positioned close to liquid, so these should be fitted with ground-fault circuit interrupters to break the circuit should any current flow to ground.

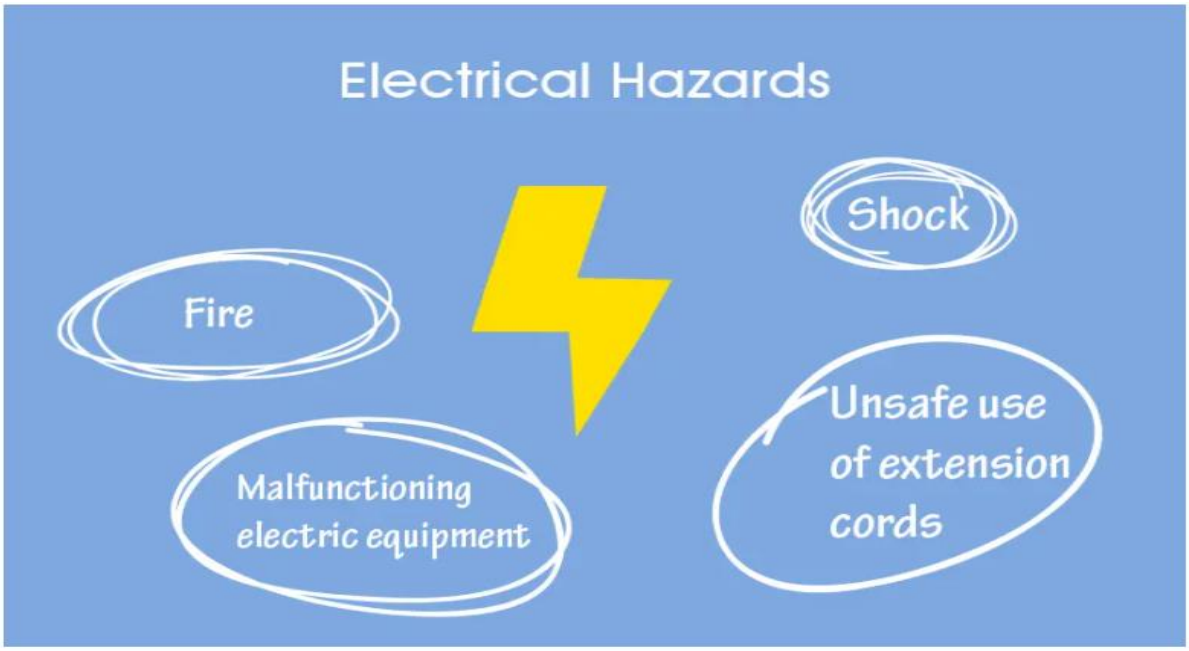
Electrical fires are another common laboratory hazard, which can occur when incorrect or unsafe cords and plugs are used. Any electrical appliances used in the laboratory should be fit for purpose, up-to-date and correspond to connected devices before they are implemented. Any electrical apparatus, from adaptors to cables, not safety-tested could compromise the safety of the lab and research team.

## Prevention

Electrical hazards can easily be overlooked in labs, which tend to have more health and safety risks than in other workplaces. Electrical hazards are potentially life threatening, however, so minimising their risk is vital. All power outlets that could be exposed to wet conditions should be equipped with ground-fault circuit interrupters. Flexible extension cords should also be well maintained and never used as a substitute for permanent wiring. Electrical pendants can be used where possible to keep cords out of the way.

## Emergency

If someone comes into contact with a live electrical source it is important not to touch them, as you could also be electrocuted. Instead turn the source off or push them clear of the source of the shock with a dry implement, such as a wooden broom handle. Your team's trained member should administer first aid. In the case of a small electrical fire, use a multi-purpose fire extinguisher – never put water on an electrical fire.




**Risk = Consequence x Likelihood**;
Where: (i) Likelihood is the Probability of occurrence of an impact that affects the environment; and, (ii) Consequence is the Environmental impact if an event occurs.

**Risk Analysis**

| | Consequence | | | | |
|---|---|---|---|---|---|
| **Likelihood** | Insignificant | Minor | Moderate | Major | Severe |
| Almost Certain | Medium | Medium | High | High | Extreme |
| Likely | Medium | Medium | Medium | High | Extreme |
| Possible | Low | Medium | Medium | High | High |
| Unlikely | Low | Low | Medium | Medium | High |
| Rare | Low | Low | Low | Medium | Medium |

## Important Items will be Observed during the experiment performance

**Apparatus or System Modules (Hardware/ Software):**
Where the main new modules to be used in the experiment are described (hardware/ software).
**Procedure:**
Leading the student on how to run the experiment. The lab work is organized in parts in order to have a clear and integrated structure of the work.
**Observations:** What particular observations are observed by the students during the experiment?

**Conclusion:**

Extracted results and concluding remarks after the experiment given by the Student.

## Learning outcomes:

What will be the learning outcome after the student has successfully completed the tasks in the Lab experiment?

## Exercises:

Extra questions and tasks for the student to carry after the lab, and include in the lab report.

## General Laboratory Procedure:

While there is no specific document to be submitted at the beginning of the Lab –unless your instructor advises you otherwise-, you are expected to read the experiment fully before you come to the laboratory.

In addition to the experiment write up, a **Lab Manual** has been prepared for every experiment. The Lab Manual is a working document, designed to help students record all lab activities (measurements, observations, answers to questions in the lab manual etc.). The student must have his/ her instructor sign the sheet before he/ she leaves.
Every student should submit a report on each experiment. The report must be self-contained, and can be read independent from the lab manual. All axes in all graphs should be clearly labeled. If there is more than one trace in the plot, they should be clearly labeled.

## Troubleshooting:

Things will not always go as expected; this is the nature of the learning process. While conducting the Experiment **think before you do anything.** If you do so you will avoid wasting time going down dead-end streets. Be logical and systematic. First, look for obvious errors that are easy to fix. Is your measuring device correctly set and connected? Are you looking at the proper scale? Is the power supply set for the correct voltage? Is the signal generator correctly set and connected? How are the variables in the code set? Is there a syntax error? And so on. Next, check for obvious misconnections or broken connections, at least in simple circuits.

As you work through your circuit, use your Lab Manual record tests and changes that you make as you go along; don't rely on your memory for what you have tried. Identify some test points in the system at which you know what the signal should be and work your way backwards from the output through the test points until you find a good signal.

## Neatness:

When you have finished for the day, return all modules to their proper storage bins, return all test leads and probes to their storage racks, return all equipment to its correct location, and clean up the lab station. If appropriate switch off the unneeded equipment. Save your files in the Computer and on any USB device for your records because you might not get the same PC System again for the next experiment. Also email your file contents to your email address as a backup.

**Answer the Following Questions:**

1) Explain the troubleshooting process.

2) Discuss the graph and its types, slope of line

3) What is mean by Accuracy, Precision

4) Discuss the Error and its types.

5) Which one is more dangerous Direct current or Alternating current? Explain.

6) Is it voltages or current that harms? Explain